



A workflow model based on parallelism for distributed organizations

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Distribution has become an increasingly common characteristic for modern service and production companies. Enterprises nowadays rely on distribution of their operations for provision of their supplies, labor, and for selling their products in dynamic global markets. Much of today enterprises efforts to cope with global markets are being directed towards the finding of effective collaboration means among their operations and partners. This research proposes a model for assisting distributed enterprises in modeling their operations by optimizing and integrating their workflow to accomplish the collaborative objective. The method developed, called the distributed parallel integration evaluation model (DPIEM) models the workflow in the distributed enterprise based on three integration scenarios. DPIEM minimizes the integrated tasks total cost by adding as many parallel servers per task as possible. The method was tested for a case of distributed assembly of two part-types. A total of eight scenarios for the case were analyzed, yielding the recommended number of parallel servers per integrated task. For comparison, each scenario was also simulated with the TIE parallel-computer environment. The TIE simulation results corroborate the DPIEM recommendation based on the lowest total cost for the case analyzed.

Keywords: Coordination, degree of parallelism, distributed assembly, distributed enterprises, distributed networks, information cost, parallel-computer simulation, supply chain, workflow integration, workflow optimization

Nomenclature

Φ	Operational cost of a integrated task graph	E	The end node of a task graph
Φ^d	Distributed operational cost	G_I	The set of task graphs in Ω
Γ	Communication cost in the integrated task graph	g_{fj}	The f -th task graph in the set of integrated task graphs (G_I)
$\pi(\nu_i)$	Execution span cost of the task ν_i	G_q	Set of d^q task graphs in ω_q
$\Psi(g_I)$	Degree of parallelism	g_{qs}	The s -th task graph in the set G_q of ω_q
Π	Execution span cost of a task graph in the set G_I	I	Integration operator
Ω	Integrated organization	I^N	Task integration operator
γ_{ij}^1	Cost for sending information from task i to j	IS_i	Input streams required for the execution of ν_i
γ_{ij}^2	Network delay cost for sending information from task i to j	K	Costs from distributed assignment of integrated tasks
γ_{ij}^3	Cost for receiving information at task j	OS_i	Output streams generated by the execution of ν_i
ν_i	A distributed or integrated task	P	Maximum number of servers available
N_I	Set of tasks in the integrated organization Ω	S	Start node initiating the execution of the task graph
ω_q	One of the r distributed organizations to integrate	S_q	Set of task graphs at ω_q
A	Average server load cost	τ_i	Reference execution cost of ν_i
		TC_i	Node cost

W	Data elements generated during execution of ν_i
X	Data elements required during execution of ν_i
Y	Data elements required and changed during execution of ν_i
Z	Data elements generated and further required during execution of ν_i

1. Introduction

Nowadays organizations display distribution of their operations as an increasingly common characteristic. Distribution has become the preferred method of industry for attaining the level of responsiveness demanded by contemporary global markets. On the other hand, coordination of remote facilities is an important aspect of distributed organizations, yet it has been the main obstacle in the development of distributed organizations supporting electronic commerce. Nevertheless, coordination has been improved thanks to better communication technologies and efficient management techniques. The advancement experienced by communication technologies has enabled distributed organizations to function in a relatively perfect and rational order by integrating the remote locations into a harmonic global operation. However, communication technology by itself does not ensure the successful integration of the distributed organization. Additional complex issues need to be addressed when designing and operating distributed organizations, among which we have detailed analysis of resource availability, process scheduling, and production planning.

The research presented here focuses on the study of task-based integration of workflow subject to constraints of information availability and tasks dependence and it resulted in the development of the distributed parallel integration evaluation model (DPIEM). The DPIEM model considers the structure, operations, and communication requirements of the distributed organization for optimizing its integrated performance and it can be applied for investigating the design of optimal integration architectures. DPIEM consists of three stages (Fig. 1). First, it integrates the distributed organization into a logical integrated operational schema. Second, it optimizes the integrated schema by parallel execution of tasks. Third, DPIEM establishes the upper boundary for the implementation cost of the parallelism recommended for operations at the integrated organization. DPIEM presents the advantage of optimizing the operating cost by implementing coordinated-parallelism in integrated operations. Current workflow integration models treat the distributed coordination problem based only on the availability of information and materials, procuring just-in-time operations. DPIEM instead takes the integration process further in terms of its applicability to a set of distributed organizations, using operational parallelism as the optimization means.

Development of DPIEM is the result from unanswered questions following the development of the parallel integration evaluation model (PIEM) (Ceroni, 1996; Ceroni and Nof, 1997), which was based on similarities observed between production and computing parallel systems (Khanna *et al.*, 1998).

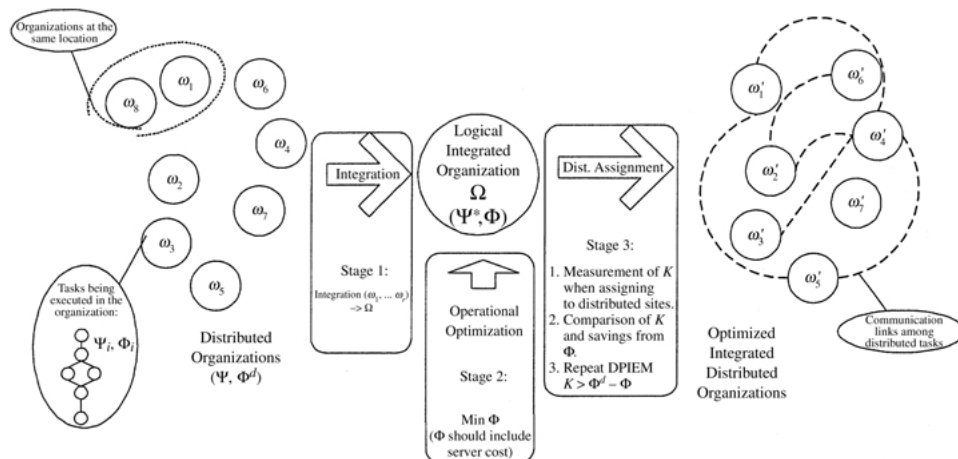


Fig. 1. Operational stages of the DPIEM model.

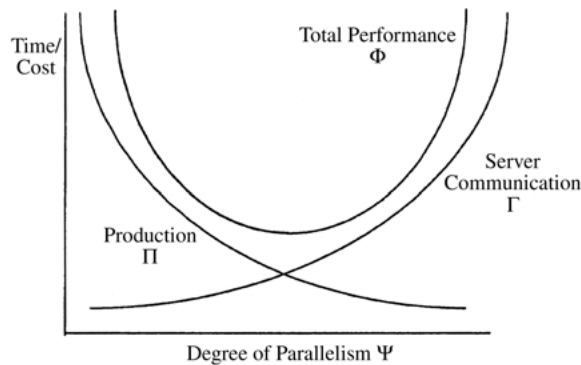


Fig. 2. Trade-off behavior between production and communication times/costs.

Outstanding similarities between parallel computing and production systems are the existence of production (goal-fulfilling) and communication (goal-delaying) times, and the trade-off behavior existing between them. As stated in Fig. 2, the production time is the running time of the servers (machines, teams, workers, or work-cells in production or computer processors in computing systems). Production time presents a trade-off with communication time (operation delay originated by communication and provision delays for servers coordination plus the congestion delays from bottlenecks in system). Cost behaves similarly relative to the production and communication cost as the degree of parallelism varies.

1.1. Significance of the problem

An increasing number of companies are basing their future on global markets. The globalization of resources and customers has shifted the focus of industrial companies from the resource control to a customer-focused control over time (Hirsch *et al.*, 1995). Competitiveness among companies nowadays relies on an increasingly important aspect: time-to-market. Pursuing shorter time-to-market imposes faster development cycles for the products while paying close attention to geographically distributed markets. In order to cope effectively in this demanding environment, companies frequently engage into collaborative partner relationships (Poirier, 1999; Armstrong and Hagel, 1996). These partnerships are invisible to customers, thanks to electronic commerce and the Internet, and allow companies to focus and coordinate their efforts for attracting customers and improving their position in

the market (Dickman, 1995; McKichan, 1990). The collaboration results in an integrated and aligned enterprise comprised of several independent companies. Partner companies combine their capabilities in generating new business opportunities to which they could not have access otherwise.

Coordination and workflow parallelism play a vital role in companies striving to operate with shorter product development cycles. Coordination allows the cooperative operation of two or more systems in the pursuit of complimentary objectives. In addition, coordination permits efficient utilization of resources and allocation of efforts in the organization. Conversely, parallelism shortens the process operational time by overlapping the execution of tasks.

1.1.1. Distributed organization

The distributed environment presents new challenges for the design, management, and operational functions in organizations. Integrated approaches for designing and managing modern companies have become mandatory practices in the modern enterprise. Historically, management had relied on a well-established hierarchy; however, the need for collaboration in modern organizations overshadows the hierarchy and imposes networks of interaction among tasks, departments, companies, etc. As a result of this interaction, three issues arise that make the integration problem critical: variability, culture, and conflicts. Variability will represent all possible results and procedures for performing the tasks in the distributed organizations. Variability is inherently present in the processes; however, distribution enhances its effects. Cultural aspects such as language, traditions, and working habits impose additional requirements for the integration process of distributed organizations. Lastly, conflicts may represent an important obstacle in the integration process. Conflicts here can be considered as the tendency to only organize and optimize locally in a dual local/global environment. Collaborative relationships, such as user/supplier, are likely to present conflicts when considered within a distributed environment. Communication of essential data and decisions plays a crucial role in allowing organizations to operate on a cooperative manner. Communication must take place in a timely basis, in order to be an effective integration facilitator and allow organizations to minimize their coordination efforts and costs.

Table 1. Data element groups in the CIMOSA Task Data Model

<i>Data group</i>	<i>Data elements</i>	<i>Information</i>
Primary	Product	Product features and its production processes, e.g., Bill of Material
	Manufacturing	Handling of orders, e.g., customer and production orders, logistic data
	planning and control	of raw materials or tools, capacity constraints
Secondary	Shop floor	Current operations of production and assembly shops, e.g., shop schedule, utilization of resources, work procedures for machine tools
	General	Company standards and guidelines, e.g., equipment, description of organization
	Responsibility	
Tertiary	Resources	

1.1.2. Coordination requirements

The organizational distributed environment has the following characteristics (Hirsch *et al.*, 1995):

- Cooperation of different (independent) enterprises;
- Shifting of project responsibilities during the product life cycle;
- Different conditions, heterogeneity, autonomy, and independence of the participants' hardware and software environments.

Based on these characteristics, the following series of requirements for the integration of distributed organizations can be established:

- Support of geographically distributed systems and applications in a multi-site production environment, and in special cases, the support of site-oriented temporal manufacturing;
- Consideration of heterogeneity of systems ontology, software, and hardware platforms and networks;
- Integration of autonomous systems within different enterprises (or enterprise domains) with unique responsibilities at different sites;
- Provision of mechanisms for business process management to coordinate the information flow within the entire integrated environment.

Therefore, coordination establishes the guidelines for the integration of the distributed organizations. These requirements are the baseline for the integration model developed in this research.

1.2. Preliminary definitions

To present the DPIEM model in the next section, some preliminary definitions are required to char-

acterize the problem of integrating distributed organizations.

Definition 1 Task (v_i)

The i -th production activity transforming input streams into output streams, and performed at a distributed or integrated organization. A task is identified by a variable (τ_i), the reference execution cost of (v_i) and two sets of attributes (IS_i , the input streams necessary to execute v_i and OS_i the output streams generated by execution of v_i). Input and output streams are organized according to a task data model (CIMOSA, 1992). The CIMOSA task data model identifies three main groups of data elements for the tasks (Table 1).

Definition 2 Distributed organization (ω_q)

A remote organization comprising a set of n tasks performed to achieve its goals. The goal may comprise a series of operational objectives, requiring the execution of subsets tasks represented by a task graph. The existence of communication links among the distributed organizations is assumed and enable their interaction to achieve integration.

$$\omega_q = \{v_i, 1 \leq i \leq n\} \quad (1)$$

ω_q represents the distributed organization after DPIEM has been applied and distributed tasks assigned.

Definition 3 Task graph (g_s)

The s -th set of nodes ($V(g_s)$) connected by a set of links or paths representing the operational sequence and communication links among the tasks ($L(g_s)$) at a distributed or integrated organization.

$$\begin{aligned}
g_s &= \{V(g_s), L(g_s)\} \\
V(g_s) &= \{N_s \cup S \cup E\} \\
N_s &= \{v_i, 1 \leq i \leq m\} (m \text{ tasks in the task graph}) \\
S &= \text{Starting node of the task graph } g_s \\
E &= \text{Ending node of the task graph } g_s \\
L(g_s) &= \{l_j, 1 \leq j \leq k\} \\
&\quad (k \text{ unidirectional edges in the task graph}) \\
V(g_s) \text{ and } L(g_s) &\text{ are finite sets.}
\end{aligned}
\tag{2}$$

The task set N_s is configured to accomplish some specific objective (e.g., production of a part or service, design of a product, establishment of a contract, etc.). Tasks in g_s are organized into sequential and parallel execution paths, determining the task execution order. Precedence relationships among tasks are determined utilizing task the dependency analysis.

Corollary

- (1) g_{qs} represents the s -th task graph in the distributed organization ω_q .
- (2) A task v_i can participate in more than one task graph and several graphs can exist in ω_q .
- (3) g_{Is} represents the s -th task graph in the integrated organization Ω .

Definition 4

Task integration process (Nof, 1994)

A process consisting of two elements of integration as follows:

$$\text{Integration} : (I, N_I) \tag{3}$$

where:

(1) I is the integration operator as defined by Papastavrou and Nof (1992), operating over tasks (I^N) in the distributed organizations ω_j and ω_k and generating tasks in Ω , such that $I: Y \rightarrow U$. Y represents the input set of tasks over which the integration operator acts. U is the result generated by the integration operator according to two scenarios considered in this research:

- (a) $I(Y_1, Y_2) \rightarrow U = Y_1 \text{ or } Y_2$
 - (b) $I(Y_1, Y_2) \rightarrow U = Y_1 \text{ and } Y_2$
- (2) N_I the set of distributed tasks being integrated.

Definition 5

Integrated organization (Ω)

The set of r integrated distributed organizations

generated by the application of the integration process to their distributed tasks as per Definition 4.

$$\Omega = \{\text{Integration } N_q\}, 1 \leq q \leq r \tag{4}$$

2. Distributed parallel integration evaluation model

The DPIEM model focuses on the optimization of the integrated operation of distributed organizations. This overall goal for the DPIEM model relates to three specific objectives: task-based integration of the distributed organizations, parallelism optimization at the integrated organization, and the boundary set for the task assignment at the distributed level. Each of these objectives relate to the result of the three modules conforming the model (Fig. 3) task integration module, integrated operation optimization module, distributed task assignment module. Each of the DPIEM modules is presented in detail in the following sections.

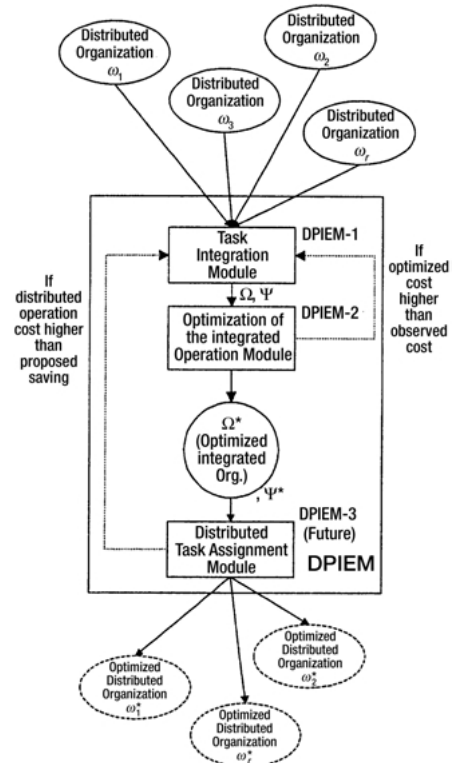


Fig. 3. Overview of DPIEM modules.

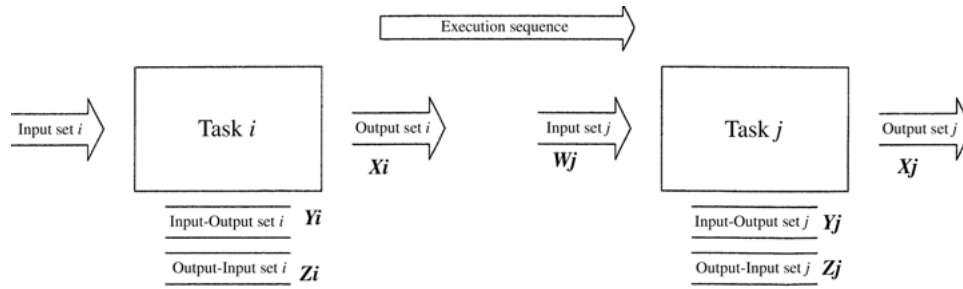


Fig. 4. Data element sets for the dependency analysis.

2.1. Task integration module DPIEM-1

The integration module first identifies and analyzes, by principles of dependency, the tasks at a local level throughout the distributed organizations. Once it devised the task graphs for the distributed organizations (ω_q), the integration module generates the set of integrated task graphs. The goal at this step is to generate the integrated organization (Ω) and its corresponding set of task graphs (G_I). Aspects such as cultural and language differences, and conflicts among the interacting distributed enterprise units can now be quantified and included in the Ω model as additional costs for the operation of the integrated organization. The integration process that generates Ω groups tasks by objective (e.g., the production of a part or component). Alternatively, the integration process may follow functional guidelines, relating tasks with similar purposes (specialization of distributed sites for example). Tasks assigned strictly to a given organization unit do not have to be included in the analysis. The set of integration-driver tasks (N^*), a sub-set of the total set of tasks (N) contains those tasks utilized for the integration process, which may affect integration-driver tasks, resulting in their change, reallocation, or elimination.

Definition 6

Relationships among task graphs

The task graph g_{qs} is part of the set of task graphs G if $V(g_{qs}) \subseteq V(G)$ and $L(g_{qs}) \subseteq L(G)$.

Definition 7

Set of distributed task graphs (G_q)

The set of task graphs representing ω_q before the integration process. The number of task graphs in G_q is d^q .

$$G_q = \{g_{qs} \mid 1 \leq s \leq d^q\} \quad (5)$$

Definition 8

Set of integrated task graphs (G_I)

The set of task graphs after the integration process has taken place. The number of task graphs in G_I is e , where:

$$G_I = \{g_{If} \mid 1 \leq f \leq e\} \quad (6)$$

The following relationship must hold true for all G_q and G_I :

$$e \leq \sum d^q \forall q. \quad (7)$$

2.1.1. Task dependency analysis

A dependency analysis method is used to find the precedence relationships among tasks at distributed and integrated level. The dependency analysis consists of the adaptation to production tasks of the dependency analysis for software procedures (Mohdaman and Evans, 1995) where data elements at the input and output of the software procedure are considered as the dependency source and follows (Bernstein, 1966) for testing the dependency. DPIEM considers production resources as machines, tools, raw materials, parts, components, assembly, operators, documents, and information. While some production resources are required for the execution of the tasks (machines, tools, or information), others are generated when the task is complete (parts, components, or information). Task data required to analyze dependencies are maintained in a data model based on the CIMOSA information structure for the enterprise activity inputs and outputs (CIMOSA, 1992 and Definition 1).

Based on the task information the test for task dependency groups task data elements in four sets (Fig. 4).

Therefore, two tasks, a and b, can be executed in

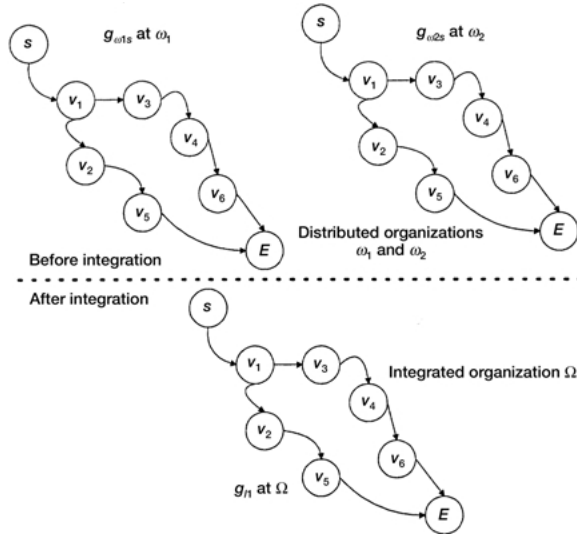


Fig. 5. Integration of identical distributed task graphs.

parallel if they do not present flow dependency and output dependencies. These two conditions can be represented as:

- (1) $(Xi \cup Yi \cup Zi) \cap (Wj \cup Yj \cup Zj) = \emptyset$ (no flow dependency from i to j)
- (2) $(Wi \cup Yi \cup Zi) \cap (Xj \cup Yj \cup Zj) = \emptyset$ (no flow dependency from i to j)
- (3) $(Xi \cup Yi \cup Zi) \cap (Xj \cup Yj \cup Zj) = \emptyset$ (no concurrent output)

For cases where the identification of resources and their utilization do not provide enough information to obtain the dependency relationships, alternative methods are proposed elsewhere in the literature. Such situations may arise when tasks present qualitative dependencies. These cases are considered out of the scope of DPIEM, however, methods as the one proposed by Kusiak and Wang (1995) may prove useful extensions to the dependency analysis considered in DPIEM.

2.1.2. Task integration

One of three basic situations can be observed, regarding parts or services produced, when integrating tasks at distributed organizations ω_1 and ω_2 : (1) identical; (2) complementary; or (3) unrelated. The proposed integration method is essentially manual and further automation needs to be analyzed and

developed. Up to this point of the analysis, only a logical representation of the integrated tasks is about to be generated. In consequence, the integration cost cannot be evaluated until the changes are performed in the distributed assignment of the tasks. Each of the three basic integration scenarios for generating the logical integrated organization of the distributed tasks is analyzed next.

(1) Situation 1: The distributed organizations produce identical parts or services. In this situation, two cases are considered:

(a) $g_{\omega_1 s} = g_{\omega_2 s} \quad 1 \leq s \leq n_1 = n_2$ (the distributed organizations perform the same processes). In this case, the integrated set of tasks will comprise the set of processes performed at either of the distributed organizations (Fig. 5):

$$g_{Is} = I(g_{\omega_1 s}, g_{\omega_2 s}) \rightarrow g_{\omega_1 s} \vee g_{\omega_2 s} \quad 1 \leq s \leq n_1 \quad (9)$$

(b) $\exists s$ such that $g_{\omega_1 s} \neq g_{\omega_2 s}$ (the distributed organizations perform at least one task that is not the same for both organizations). Two alternatives considered are modifiable and non-modifiable tasks.

(i) Modifiable tasks

$$g_{Is} = I(g_{\omega_1 s}, g_{\omega_2 s}) \rightarrow g_{\omega_1 s} \text{ iff } N_{\omega_1 s} \leq N_{\omega_2 s} \times 1 \leq s \leq n_1 \quad (10)$$

In this case, the distributed tasks of either distributed organization can be modified. The distributed organization with lower number of tasks in its task graphs is selected as the integrated task model. This approach imposes changes to the processes being performed at the non-selected organization. The cost of the proposed changes must be analyzed and compared to the expected reward as result of the integration. An example of this case is presented in Fig. 5.

(ii) Non-modifiable tasks

$$g_{Is} = I(g_{\omega_1 s}, g_{\omega_2 s}) \rightarrow U = \{g_{\omega_1 s} \cup g_{\omega_2 s}\} - \{g_{\omega_1 s} \cap g_{\omega_2 s}\} \quad 1 \leq s \leq \max\{n_1, n_2\} \quad (11)$$

In this case, the distributed tasks cannot be modified and a joint logical representation of the production processes at each organization is created. Common processes are represented only once, being shared by more than one task graph if necessary (Fig. 6).

In Fig. 7, Ω contains the task graphs of organization ω_1 and ω_2 . Task v_3 drives the integration process in this case and it is represented only once in the

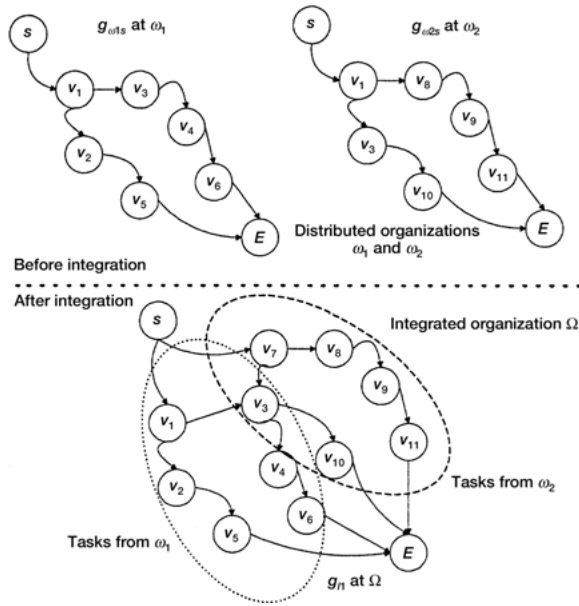


Fig. 6. Integration of different processes producing identical part-types.

integrated task graph. The rest of the distributed tasks are not modified but taken directly into the integrated task graph.

(2) Situation 2: The distributed organizations produce complementary components or services. This case requires coordination among the organizations to produce the final assembly part. The set of integrated task graphs constitutes all the task graphs at the distributed organizations:

$$g_{Is} = I(g_{\omega_1s}, g_{\omega_2s}) \rightarrow U = \{g_{\omega_1s} \cup g_{\omega_2s}\} \quad 1 \leq s \leq \max\{n_1, n_2\} \quad (12)$$

The basic information to consider in the integration of the distributed task graphs is the sequence of the assembly or service produced at each organization (Fig. 7).

(3) Situation 3: The distributed organizations generate different parts or services. In this scenario, the execution sequence of tasks between the distributed organizations is irrelevant.

$$g_{Is} = I(g_{\omega_1s}, g_{\omega_2s}) \rightarrow U\{g_{\omega_1s} \cup g_{\omega_2s}\} \quad 1 \leq s \leq \max\{n_1, n_2\} \quad (13)$$

The integration process is based solely on the dependency of inputs and outputs among tasks in the distributed organizations. Here, integration is led

by finding tasks that require the same input resources (machines, materials, etc.). Alternatively, integration can also be implemented with tasks generating similar intermediate components. The task graphs presenting tasks with common inputs or outputs are shown in Fig. 8.

2.2. Task parallelism optimization module DPIEM-2

This module optimizes Φ , the total cost of Ω . The optimization model is based on the concepts outlined in (Ceroni, 1999; Ceroni and Nof, 1997; Qin *et al.*, 1991) and it is an adaptation of the optimization model in (Ramaswamy *et al.*, 1997) for parallel processor allocation. The DPIEM optimization model minimizes the maximum between the operational cost and the average server load for the integrated task graph. These two cost elements complement each other in determining the total cost for the integrated tasks. While searching for the lowest operational cost, the system may require a very high number of parallel servers in the critical path as well as for the rest of the integrated tasks. This situation is detected by the average server load cost. The optimization problem formulated becomes non-linear and requires the approximation of its solution to integer values. The complexity of the optimization model is $O(n^{2.5})$ (Ramaswamy *et al.*, 1997).

Definition 9

Communication cost (Γ)

The non-operational cost (tasks execution delay) for the execution of the set of tasks in G_I . The communication cost is computed for every edge representing exchange of information, parts, materials, or tools among the tasks. It is assumed that there is no difference between inter-site and within site communication costs. It must be noted that adjustments to the communication cost parameters will be necessary for reflecting the sharing of the transportation media by the tasks, its delivery speed, and its capacity. In this research, only data transmission is considered for evaluating the communication cost.

The communication cost is considered by three components: sending the information at the source task (γ^1), network delay (γ^2), and receiving the information at the target task (γ^3). The cost associated to each of these components is as given in Equations 14 to 17.

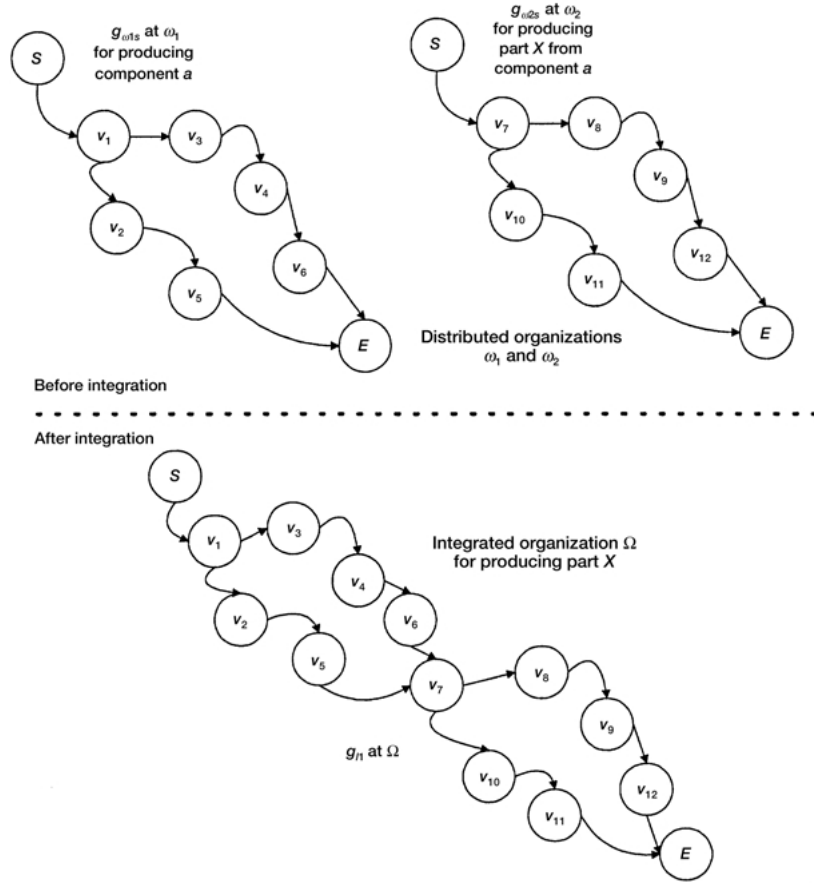


Fig. 7. Integration of task graphs for producing components of part-types.

$$\Gamma = \gamma_{ij}^1 + \gamma_{ij}^2 + \gamma_{ij}^3 \quad (14)$$

$$\gamma_{ij}^1 = \max\left(1, \frac{p_j}{p_i}\right) * tc_{ss} + L * \left(\frac{1}{p_i}\right) * tc_{ps} \quad (15)$$

$$\gamma_{ij}^2 = \frac{L * tc_n}{p_i * \max\left(1, \frac{p_i}{p_j}\right)} \quad (16)$$

$$\gamma_{ij}^3 = \max\left(1, \frac{p_i}{p_j}\right) * tc_{sr} + L * \left(\frac{1}{p_j}\right) * tc_{pr} \quad (17)$$

where:

L is the length in bytes of the information message;
 tc_{ss}, tc_{ps} are the startup and per byte transmission cost for sending the data from task i ; tc_n is the network cost per message byte; tc_{sr}, tc_{pr} are the startup and per byte receiving cost at task j ; and p_i, p_j are the number of parallel servers at tasks i and j , respectively.

Definition 10

Task operational cost (π_i)

Execution cost of each task, computed per Amdahl's law, which states the execution cost of a task has two components, one serial and another parallel. The serial component must be executed by one server, while the parallel component can be executed by two or more servers. Let $\alpha_i (0 \leq \alpha_i \leq 1)$ represent the serial part of the task, then π_i is as given by Equation 18 with τ_i is the completely serial task execution cost (when executed by one server).

$$\pi_i = \left(\alpha_i + \frac{1 - \alpha_i}{p_i}\right) * \tau_i \quad (18)$$

τ_i includes a contributing factor related to the reliability of the servers. It is assumed that bad

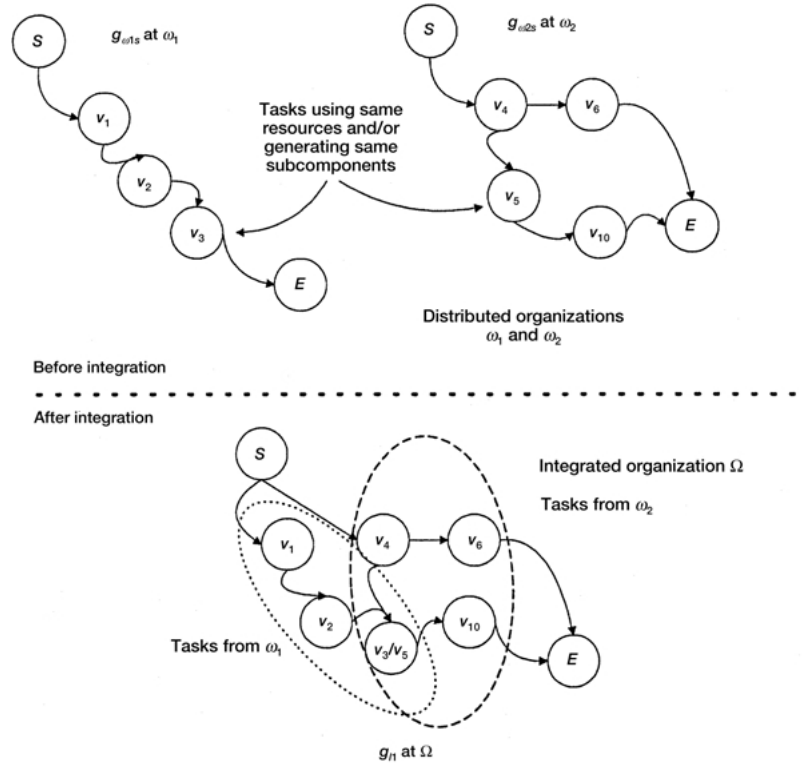


Fig. 8. Integration of task graphs for producing different part-types.

reliability of servers will result in higher costs due to the utilization of replacement servers, repairs, and related delays.

Definition 11

Operational cost (Π)

The cost for the execution of the set of tasks in the integrated task graph G_I . The operational cost for the task graph is computed by adding the task execution cost of the tasks in the task graph's critical path:

$$\Pi = \sum_{\text{Critical path}} (\pi_i + \max_{\text{Predecessors}} (\gamma_{pi}^2)) \quad (19)$$

The expression to compute the operational cost does not consider that a large amount of servers could end up assigned to non-critical tasks, distorting the cost measurement. To remedy this situation, additional cost elements are considered.

Definition 12

Node cost per server (TC_i)

The task based cost comprising the task execution cost, the cost for each server assigned, and the communication cost to preceding and succeeding

tasks. The node cost per server is computed for each task in the task graph as:

$$TC_i = \sum_{p \in \text{Predecessors}} \gamma_{pi}^1 + \pi_i + PC_i + \sum_{s \in \text{Successors}} \gamma_{is}^3 \quad (20)$$

In Equation 20, PC_i is the utilization cost per server assigned to task i . TC_i is independent of the task graph critical path and its objective is to evaluate the global cost for the tasks in the task graph.

Definition 13

Average server load cost (A)

Cost measurement of the average processor-load cost for the integrated task graph. The average server load cost is defined as:

$$A = \frac{1}{P} * \sum_{i=1}^{N_I} (p_i * TC_i) \quad (21)$$

Definition 14

Degree of parallelism (Ψ)

A function over a task graph yielding its maximum number of concurrent (with parallel execution)

servers. This index provides more information than simply the number of servers when considering alternative parallel implementations for the system. Ψ can be calculated for distributed and integrated task graphs. Ψ^* will denote the optimal degree of parallelism of G_I , the set of integrated task graphs.

To clarify the previous definitions, consider the task graph and parameters in Fig. 9.

Calculation of Π , TC_i , and A follows the expressions in Equations 20 to 21. Assuming initially one server per task, the calculation is as follows:

$$\begin{aligned}\Pi &= \pi_1 + \pi_4 + \gamma_{14} + \pi_6 + \gamma_{46} = 10 \\ &+ 10 + 10 + 10 + 10 = 50 \text{ \$/Server}\end{aligned}$$

since,

$$\begin{aligned}\pi_1 &= \pi_4 = \pi_6 = \left(\frac{\alpha_1 + (1 - \alpha_1)}{p_1} \right)^* \\ \tau_1 &= \left(\frac{0.2 + (1 - 0.2)}{1} \right) * 10 = 10 \text{ \$/Server},\end{aligned}$$

and

$$\begin{aligned}\gamma_{14}^2 &= \gamma_{46}^2 = \frac{L * tc_n}{(p_1 * \max(1, p_4/p_1))} \\ &= \frac{2000 * 0.005}{(1 * 1)} = 10 \text{ \$/Server}\end{aligned}$$

The node cost (TC_i) instead requires its computation for all nodes in the task graph. For node number 1, TC_1 is calculated as follows:

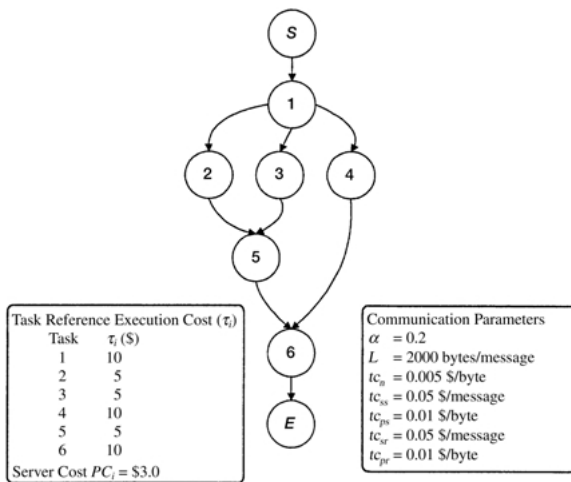


Fig. 9. Task graph and parameters for the illustration case.

$$\begin{aligned}TC_1 &= \pi_1 + PC_1 + \gamma_{12}^3 + \gamma_{13}^3 + \gamma_{14}^3 \\ &= 10 + 3 + 20.05 + 20.05 + 20.05 \\ &= 73.15 \text{ (\$/Server)}\end{aligned}$$

since

$$\begin{aligned}\gamma_{12}^3 &= \gamma_{13}^3 = \gamma_{14}^3 = \max(1, p_1/p_2) * tc_{sr} + L * (1/p_2) \\ &* tc_{pr} = 1 * 0.05 + 2000 * 0.01 \\ &= 20.05 \text{ \$/Server}\end{aligned}$$

Finally, the Average Server Load is calculated with the results for TC_i for all six tasks as:

$$A = (1/P) * \sum (p_i * TC_i) = 57.28 \text{ \$/Server}$$

where P is assumed equal to 6 (one dedicated server per task).

In the previous calculations, the value of A is greater than Π , and therefore it dominates the optimization. However, when additional servers are included in the task graph, the values for A and Π change accordingly. Table 2 shows the shift from A to Π when the number of servers at each task is changed from 1 to 15 (same number of servers for each task). The shift observed in the dominant variable ($\Psi = 21$) changes the focus of the optimization problem, where the optimal number of servers will be determined by the minimum value of Π instead of A .

Definition 15

Integrated total cost (Φ)

The maximum between the operational cost for the

Table 2. Changes in Π and A for different values of Ψ

Servers/task	Ψ	Π	A
1	3	50.00	57.28
2	6	28.00	30.95
3	9	20.67	22.17
4	12	17.00	17.78
5	15	14.80	15.15
6	18	13.33	13.39
7	21	12.29	12.14
8	24	11.50	11.20
9	27	10.89	10.47
10	30	10.40	9.88
11	33	10.00	9.41
12	36	9.67	9.01
13	39	9.39	8.67
14	42	9.14	8.38
15	45	8.93	8.13

integrated task graph (Π) and the average server load cost (A).

$$\Phi = \max(\Pi, A) \quad (22)$$

Both cost expressions for Π and A in Equation 22 include the number of servers, allowing the determination of the number of servers for achieving the minimum operating cost. The focus of both cost expressions however, is different. Π emphasizes the set of tasks comprising the critical path in the integrated task graph, while A on the other hand, emphasizes the cost from all servers operating in the integrated organization. Φ is minimized by the next non-linear, real optimization problem.

(1) Minimize $\Phi = \max(\Pi, A)$

subject to:

$$1 \leq p_i \leq P \quad i = 1, n$$

(2) $p_i^* = \lfloor p_i \rfloor \quad i = 1, n$

p_i , the decision variable of the optimization problem represents the number of servers assigned to each integrated task. The values obtained are approximated to the closest lower integer value since the optimization problem maximizes the number of servers. P is the maximum number of servers available for executing the tasks in the integrated task graph. Handling of P , as well as the server cost, provides a means to test different global configurations of servers. Scalability analysis for the integrated organization is possible by adjusting P to different values, and then solving the optimization problem for each of them. If no additional constraints are to be imposed on the solution of the optimization problem, P can be set to a practically large value. Proof of the existence of an optimal solution for the optimization problem can be found in (Ceroni, 1999).

Definition 16

Distributed operational cost (Φ^d)

The cost derived from the operation of the distributed organizations (ω_i) at the same activity level and capacity of the integrated organization (Ω). This cost is used as comparative measurement for the improvement (or penalty) obtained with the optimized integrated operational cost.

2.3. Distributed task assignment module DPIEM-3

The purpose of this module is to set the upper boundary for the costs resulting from assigning the integrated tasks to their distributed location.

Definition 17

Distributed task assignment cost (K)

The costs derived from the distributed assignment of the integrated tasks. This stage has markedly practical-operational characteristics, however, it is important to validate the feasibility at this stage of the DPIEM proposed operational schema. These additional costs during the assignment stage were not considered during the optimization stage (DPIEM-2) and they may arise from two situations, reallocation or modification of tasks.

Another source of additional cost, difficult to quantify before the assignment stage, is the complex interaction existing among distributed tasks. Some examples of complex interactions are dynamic economic conditions inhibiting task integration, and barriers to an effective task integration (e.g., conflicts, language, culture, etc.). The departure from the predicted operational cost performance by the DPIEM model must be monitored in order to ensure the applicability of the proposed solution. Radical differences will illustrate that the assumed model conditions at distributed and local levels are no longer valid, needing careful revision to apply DPIEM again.

3. Experimentation with the model

The experimentation with the model is presented in two parts. First, a base scenario is proposed and several scenarios generated from it, some generated with DPIEM and others manually generated (Table 3). Second, the scenarios are simulated in the TIE computer environment (Khanna and Nof, 1994) recoding their processing and communication costs to analyze the effectiveness of DPIEM to predict the lowest total cost design for the system. The optimization of the integrated operation was performed using the software GAMS for non-linear optimization problems. The parameters used to apply DPIEM are as presented in Table 4. The data transmission parameters of Table 4a were established per the communication parameters used in (Ceroni *et al.*, 1999), the modeling by Lin and Prasanna (1995), and the experiments by Xu and Hwang (1996). The server cost was set at a relatively low value (\$ 3/server) to avoid constraining the model in using large number of servers. Parameters for the task operational cost (Table 4b) assume a normal distribution with the mean and standard deviation as shown.

Table 3. Scenarios in the experimental investigation of DPIEM

Scenario	Specifications					
	<i>Ass'y Model</i>	<i># Sites</i>	<i>α(%)</i>	<i># Mfg Servers/Site</i>	<i>Ass'y Servers/Site</i>	<i>ψ</i>
Base scenario	Other	3	20	2	2	6
Scenario 1 (DPIEM)	Geometric	2	20	12	8	24
Scenario 2 (DPIEM)	Geometric	2	60	12	8	24
Scenario 3 (DPIEM)	Geometric	4	20	6	4	24
Scenario 4 (DPIEM)	Geometric	4	60	6	4	24
Scenario 5	Geometric	4	60	8	4	32
	<i>Ass'y Model</i>	<i># Sites</i>	<i>Mfg—Ass'y Operation</i>	<i># Mfg. Servers/Site</i>	<i># Ass'y Servers/Site</i>	<i>ψ</i>
Scenario 6	Combinatorial	2	Sequential	2	6	12
Scenario 7	Combinatorial	2	Integrated	2	6	12
Scenario 8 (DPIEM)	Combinatorial	2	Integrated	3	3	6

Table 4. Parameters for the experimental investigation of DPIEM

(a) DPIEM communication cost model parameters

<i>Parameter</i>	<i>Description</i>	<i>Value</i>
L	Message length	2000 (bytes)
tc_n	Per byte transmission cost on network	0.005 (\$/byte)
tc_{ss}	Startup cost at sending task	0.05 (\$/message)
tc_{ps}	Per byte injection cost at sending task	0.005 (\$/byte)
tc_{sr}	Startup cost at receiving task	0.05 (\$/message)
tc_{pr}	Per byte retrieving cost at receiving task	0.005 (\$/byte)
CP_i	Server cost (same for all servers)	3.00 (\$/server)

(b) TIE simulation parameters

<i>Parameter</i>	<i>Description</i>	<i>Value</i>
μ_e	Mean task operational cost	2.0 (\$)
σ_e	Std. deviation of the task operational cost	0.4 (\$)
μ_f	Mean of task failure plus repair cost	0 (\$)
σ_f	Std. deviation of task failure plus repair cost	$0.01 * \sigma_e$ (\$)

3.1. Experiment with distributed assembly

Two part-types, A and B, are manufactured and assembled at a number of distributed sites. Each part-type is represented by one of the assembly models:

Geometric and Combinatorial, as described in (Nof *et al.*, 1997). The assembly model and number of components for the part-type determine how long it takes to manufacturing and assembly servers to finish each unit being ordered. Coordination is needed because sub-assemblies of both part-types are produced at different sites from where the final assembly takes place. Both manufacturing and assembly tasks are assumed available at each site, however, assembly tasks are assumed not movable. Capacities at each distributed site are large enough for satisfying the system demand for parts. Scenarios are proposed for measuring the effect of changes in the assembly model, the number of sites, serial content of manufacturing and assembly tasks, and coordination mode of the manufacturing and assembly tasks.

3.2. Base scenario

The base scenario comprises the distributed operation of three sites for producing part-types A and B. Sites 2 and 3 produce part-type B, while site 1 produces part-type A. Part-type A has 8 components while part B has six components, assembled according to Fig. 10. Orders for both part-types arrive at one of the sites, which distributes the requirements to the other sites. Orders received for part-type B are distributed equally for sites 2 and 3. Each site has two manufacturing servers and two assembly servers operating sequentially (manufacturing and then assembly) to produce the part-types.

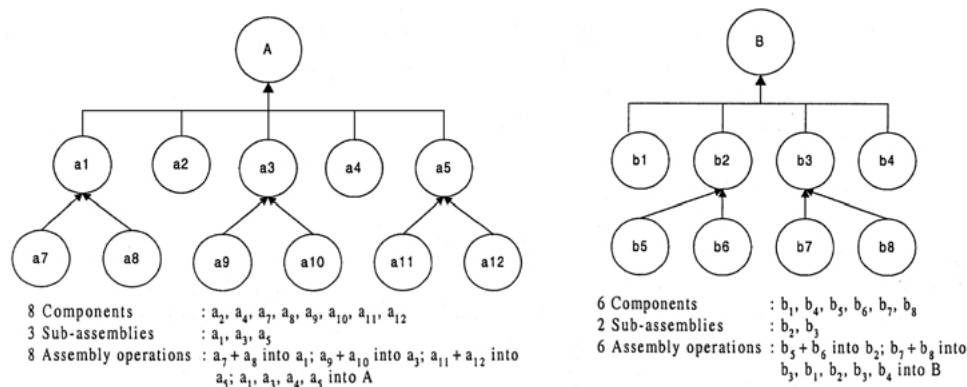
Table 5. Task data model for the experiment

Primary information	Secondary information	Tertiary information	Dependency information	
			Task	Dependency type
Task: Manufacturing of components				
●Component technical and processing specifications	Manufacturing task responsible	Mfg. server(s) personnel	Assembly task	Flow (Manufacturing to Assembly)
●Shop floor status		Tooling Materials		
●Mfg. planning and control				
●Operation mode (sequential or integrated)				
●Interaction with manufacturing tasks at other sites				
Task: Assembly of components and subassemblies				
●Assembly, subassembly specifications	Assembly task responsible	Assembly server(s)	Manufacturing task	Flow (Manufacturing to Assembly)
●Shop floor status		Personnel Tooling/fixtures Testing equipment		
●Assembly planning and control				
●Assembly model				
●Operation mode (integrated or sequential)				
●Interaction with assembly tasks at other sites				

3.2.1. Integration in base scenario

In order to establish the relationship among the distributed tasks, a dependency analysis is performed over the manufacturing and assembly tasks in the distributed system. No difference is assumed between assembly of components and subassemblies. Besides

the manufacturing-assembly sequence, sites also must interact for distribution of customer orders and requesting unused capacity at other sites. In all scenarios, the dependency analysis considers mainly the flow dependency existing between manufacturing and assembly tasks per the data in Table 5. In

**Fig. 10.** Assembly models for part-types A and B in the base scenario.

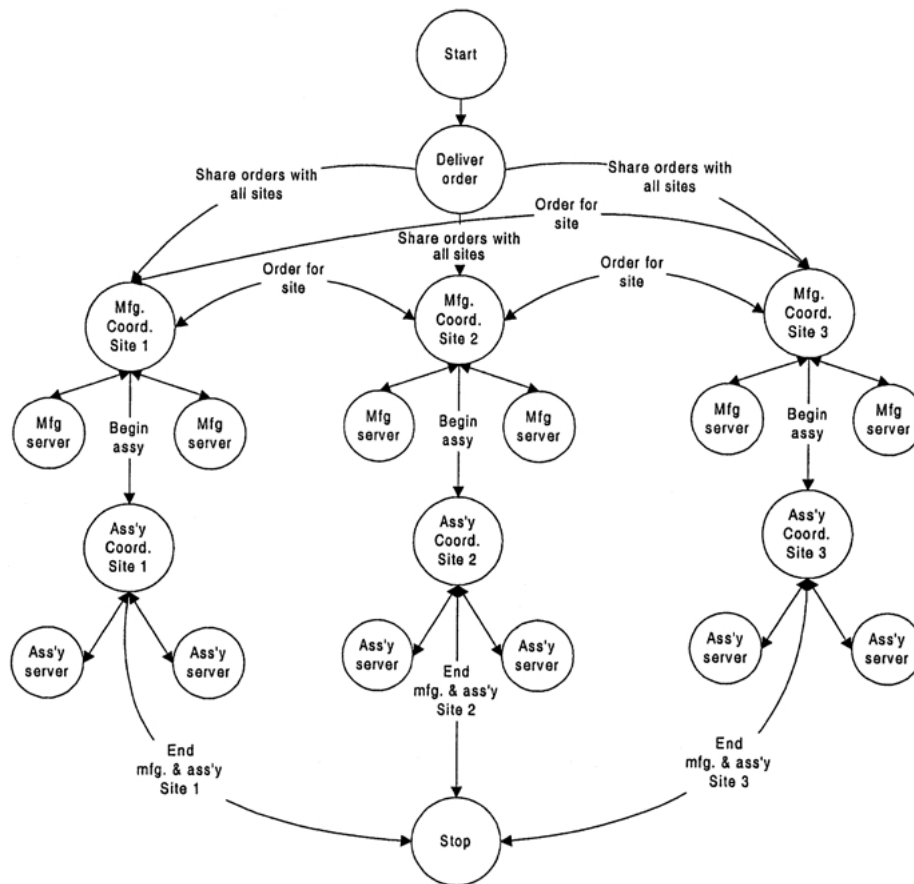


Fig. 11. Task graph for the base scenario.

scenarios 1 to 6 in Table 3, operation is assumed sequential (first manufacturing and then assembly). Scenarios 7 and 8 implement a concurrent operation where the execution of manufacturing and assembly tasks are interleaved with part-types being produced gradually along the production run.

Figure 11 shows the task graph of the base scenario, which was simulated with 5 replications and their results recorded. These results will be compared to the rest of the scenarios.

All scenarios in the experiment are based on modifications introduced to the base scenario. The modifications generating scenarios 1 to 4, and 8 relate to the use of DPIEM and three specific improvements: (1) Consolidation of sites 2 and 3 for producing part-type B; (2) Utilization of Geometrical and Combinatorial assembly models, and (3) Integration of manufacturing and assembly to reduce production cycle times. The rest of the scenarios were designed in order to compare the performance of DPIEM.

3.3. Scenarios with geometric assembly model

The part-types in these scenarios follow a Geometric Assembly model (Fig. 12). The experiment in here is organized into 5 scenarios comprising different number of sites and values of the serial job content for the manufacturing and assembly tasks. Application of the Geometric assembly model implies the redesign of the part-types, regarding components, manufacturing, and assembly processes.

3.3.1. DPIEM-1: Integration of DPIEM scenarios with geometric assembly model

Scenarios 1 and 2 of the experiment were defined with two sites, each of them produces one part-type (part-type A in site 1 and part-type B in site 2). The dependency analysis of manufacturing and assembly tasks is based on information in Table 5. Its result consists in the sequencing first manufacturing and then assembly tasks at each site (based on flow

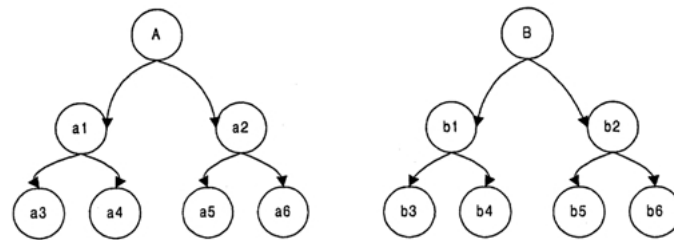


Fig. 12. Geometric assembly model for scenarios 1 to 5.

dependency of manufacturing and assembly tasks). The integrated task graph for distributed sites is shown in Fig. 13. The number of servers shown in Fig. 13 corresponds to the result of the optimization stage of DPIEM and it has been included here for complementing the graph information. Four sites

are considered in scenarios 3 to 5. For these scenarios, the sites are dedicated to the manufacturing and assembly of part-types according to the assignment shown in Table 6. Similar task graphs developed for scenarios 3 to 5 are omitted due to lack of space.

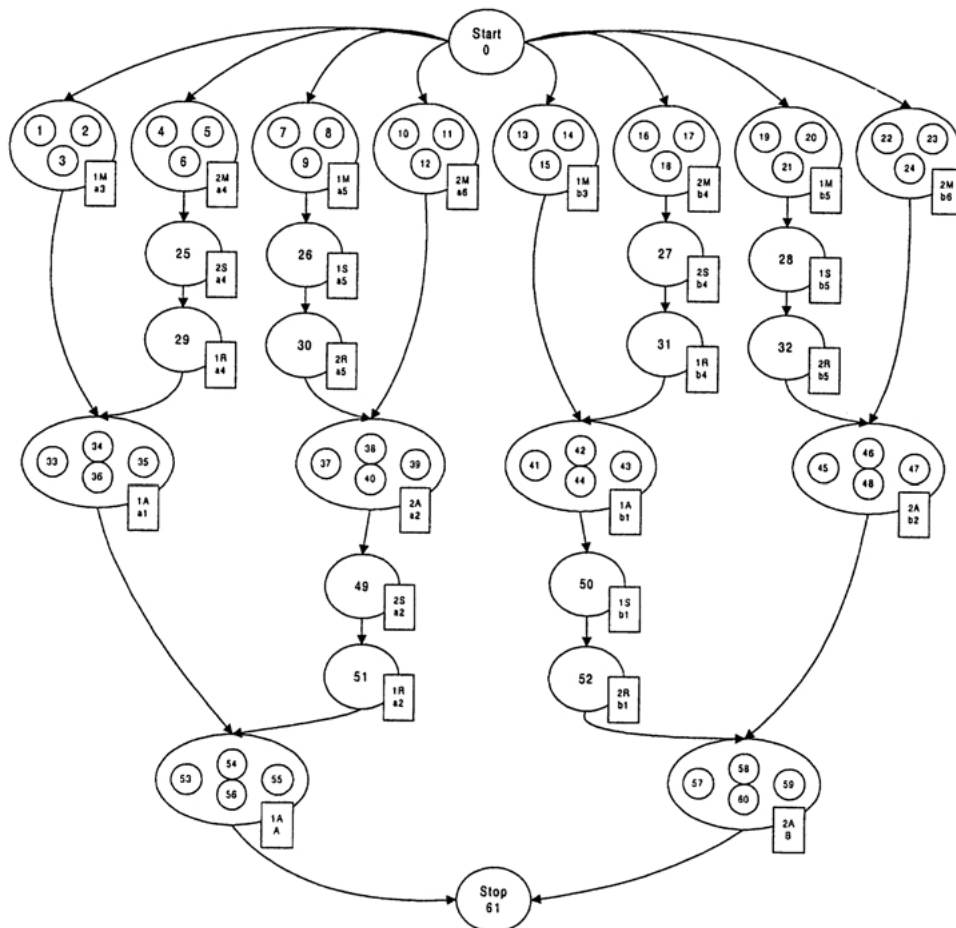


Fig. 13. Task graph for scenarios 1 and 2 (12 manufacturing and 8 assembly servers per site, $\psi^* = 24$ Legend: ellipses: parallel servers, circles: servers, rectangles: site-activity-component/part, Activity: M: manufacturing, A: assembly, S: sending, R: receiving).

Table 6. Assignment of components and part-types to sites in scenarios 3 to 5

Site	Manufactures	Assembles
1	a3, b3	a1, a
2	a4, b4	a2, b
3	a5, b5	b1
4	a6, b6	b2

Table 7. Optimal cost performance measurements (before approximation to its integer solution)

Scenario	Φ (\$/server)	Π (\$/server)	A (\$/server)
1	20.248	8.148	20.248
2	20.258	8.196	20.258
3	24.758	10.405	24.758
4	24.767	10.453	24.767

3.3.2. DPIEM-2: Optimization of DPIEM scenarios with geometric assembly model

For each scenario the optimization model was formulated with the results shown in Tables 7, 8, and 9. The optimization model was solved using the parameters in Table 4 and the GAMS (General Algebraic Modeling System) software on a Unix/Solaris platform. The cost performance measurements for scenarios 1 to 4 as obtained with the optimization model are presented in Table 7.

3.3.3. Comparison: TIE simulation of the scenarios with geometric assembly model

Once we establish the design, the scenarios were simulated in the TIE computer environment. The computer simulation of the manufacturing and assembly tasks is based on the parallel execution of programs exchanging information allowing their synchronization. Experiments were simulated with 5 replications to compensate for the effect of the randomness in the generation of task processing times and failure plus repair delays (task processing cost). Simulation results for scenarios 1 to 4 are presented in Table 10 and statistically analyzed in Section 3.6.

3.4. Scenarios with combinatorial assembly model

In experiments of scenarios 6 to 8 of the Distributed Assembly Case, two part-types are manufactured and assembled at two sites. Testing with the Combinatorial

Table 8. Optimal number of manufacturing and assembly servers per site for scenarios 1 and 2 ($\psi^* = 24$)

Task	Optimal number of servers	Integer number of servers
1Ma3: Mfg. Of a3 at site 1	3.855	3
2Ma4: Mfg. Of a4 at site 2	3.849	3
1Ma5: Mfg. of a5 at site 1	3.883	3
2Ma6: Mfg. of a6 at site 2	3.885	3
1Mb3: Mfg. of b3 at site 1	3.885	3
2Mb4: Mfg. of b4 at site 2	3.883	3
1Mb5: Mfg. of b5 at site 1	3.849	3
2Mb6: Mfg. of b6 at site 2	3.855	3
1Aa1: Ass'y of a1 at site 1	4.711	4
2Aa2: Ass'y of a2 at site 2	4.733	4
1Ab1: Ass'y of b1 at site 1	4.733	4
2Ab2: Ass'y of b2 at site 2	4.711	4
1AA: Ass'y of A at site 1	4.735	4
2AB: Ass'y of B at site 2	4.735	4

Assembly model is organized in three scenarios as described in Table 11. Two changes are tested with DPIEM. First, implementation of Combinatorial Assembly for both part-types, which increases the number of components and eliminates sub-components (a part is finished in one cycle assembly if 6 servers are available). Second, the integration of manufacturing and assembly tasks is implemented by performing alternate cycles of manufacturing and

Table 9. Optimal number of manufacturing and assembly servers per site for scenarios 3 and 4 ($\psi^* = 24$)

Task	Optimal number of servers	Integer number of servers
1Ma3: Mfg. of a3 at site 1	3.855	3
2Ma4: Mfg. of a4 at site 2	3.849	3
3Ma5: Mfg. of a5 at site 3	3.855	3
4Ma6: Mfg. of a6 at site 4	3.855	3
1Mb3: Mfg. of b3 at site 1	3.859	3
2Mb4: Mfg. of b4 at site 2	3.859	3
3Mb5: Mfg. of b5 at site 3	3.853	3
4Mb6: Mfg. of b6 at site 4	3.855	3
1Aa1: Ass'y of a1 at site 1	4.711	4
2Aa2: Ass'y of a2 at site 2	4.692	4
3Ab1: Ass'y of b1 at site 3	4.669	4
4Ab2: Ass'y of b2 at site 4	4.708	4
1AA: Ass'y of A at site 1	4.176	4
2AB: Ass'y of B at site 2	4.702	4

Table 10. Simulation results for scenarios 1 to 5

Scenario	Φ (\$/server)					Π (\$/server)				
	Replication					Replication				
	1	2	3	4	5	1	2	3	4	5
Base	4.191	5.863	3.078	3.925	4.194	3.073	4.196	2.393	3.462	3.062
1	0.650	0.650	0.632	1.022	0.769	0.065	0.065	0.051	0.063	0.051
2	1.411	0.893	0.655	0.645	0.686	0.111	0.114	0.108	0.102	0.106
3	0.937	1.019	1.034	1.093	1.030	0.050	0.058	0.041	0.063	0.064
4	0.996	1.047	1.000	0.984	0.991	0.110	0.088	0.136	0.106	0.107
5	1.309	3.258	1.318	1.307	1.311	0.076	0.077	0.076	0.071	0.069

assembly (the number of cycles will depend on the number of servers available). The first two scenarios (6 and 7) are determined manually and have 2 manufacturing and 6 assembly servers. The number of servers was established without using DPIEM and they are intended to serve as reference to the DPIEM-based scenario 8. The number of sites is maintained fixed at two sites. The base scenario for these experiments is the same as utilized for scenarios 1 to 5.

3.4.1. DPIEM-1: Integration of DPIEM scenarios with combinatorial assembly model

The integration process is based on the integration situation 3 (production of different parts). The task

graph for scenarios 6 and 7 is presented in Fig. 14 including the number of servers per the optimization stage. Similarly generated, task graph of scenario 8 is omitted here due to lack of space.

3.4.2. DPIEM-2: Optimization of DPIEM scenarios with combinatorial assembly model

Following the optimization model (Section 2.2), the parameters of Table 4, and the additional parameters of Table 12, the optimal number of servers (Table 13) for the tasks in scenario 8 are determined. The optimization model is solved by implementing the model in the software GAMS. The cost performance measurements for the optimization of scenario 8 are $\Phi = 21.612$; $\Pi = 9.162$; and $A = 21.612$.

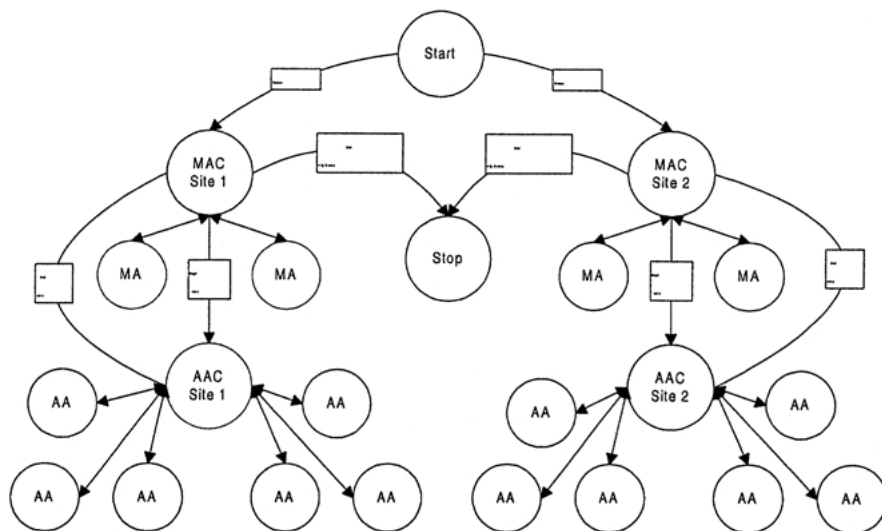


Fig. 14. Task graph for scenarios 6 and 7 (2 manufacturing servers, 6 assembly servers, and sequential operation. Legend: MAC: manufacturing coordinator, MA: manufacturing server, AAC: assembly coordinator, AA: assembly server).

Table 11. Scenarios of the distributed assembly experiment with combinatorial assembly model

Scenario	Mfg. servers/site	Ass'y servers/site	Number of sites	Operational mode	Ψ
6*	2	6	2	Sequential	12
7*	2	6	2	Integrated	12
8	3	3	2	Integrated	6

*Manually generated scenarios.

Table 12. Additional DPIEM parameters for scenario 8

Parameter	Value
τ_i : Task reference execution cost	\$ 2.0 (equal for all tasks)
αi : Serial content of tasks	0.2 (equal for all tasks)

Table 13. Optimal number of manufacturing and assembly servers in scenario 8

Task	Optimal number of servers	Integer number of servers
Mfg. of components for A	3.866	3
Mfg. of components for B	3.866	3
Ass'y of components for A	3.883	3
Ass'y of components for B	3.883	3

3.4.3. Comparison: TIE simulation of the scenarios with combinatorial assembly model

The TIE simulation results for scenarios 6 to 8 are presented in Table 14 and statistically analyzed in Section 3.6.

3.5. DPIEM-3: Distributed assignment of the integrated tasks

The upper bound for the cost of the distributed assignment of the task need to be compared with the savings expected from the operation of the new integrated organization. For this experimental investigation, however, all we can establish is the resulting maximum cost of the distributed assignment. In situations where the assignment cost is foreseen as higher than the reward from applying the recommendation from DPIEM model, the three steps of the methodology should be carefully reviewed checking the total cost for sources of additional savings. For all the scenarios of the case, the savings generated and the maximum allowed cost for the distributed task assignment process is presented in Table 15.

3.6. Analysis of experiment results

The results from the TIE simulation experiments are analyzed first by statistical tests. Next, the statistical tests are interpreted, extracting the guidelines of the DPIEM application and general conclusions on the DPIEM development and its applicability. As a first approach to the investigation of the impact of the DPIEM application, the results of Table 16 are useful for determining general hypotheses of the DPIEM

Table 14. TIE simulation results for scenarios 6 to 8

Scenario	Φ (\$/server)					Π (\$/server)				
	Replication					Replication				
	1	2	3	4	5	1	2	3	4	5
Base	4.191	5.863	3.078	3.925	4.194	3.073	4.196	2.393	3.462	3.062
6	1.764	2.360	1.994	1.530	1.845	1.161	1.604	1.301	1.039	1.280
7	1.993	2.403	2.470	2.077	1.640	1.335	1.494	1.753	1.172	1.062
8	2.769	2.738	2.823	1.989	2.548	1.966	1.523	1.888	1.309	1.765

Table 15. Maximum K for scenarios of the case (base scenario average $\Phi^d = 4.250$)

Scenario	K (\$/server)
1	3.505
2	3.392
3	3.227
4	3.246
5	2.549
6	2.351
7	2.133
8	1.677

performance. The metrics related to the designs generated by DPIEM for each scenario where applied are presented in Table 17.

In order to investigate the details of the results of Table 16, statistical analysis of the experiment results is performed next. The tests consist of analysis of variance and difference of means. The tests will provide the information required for inferring general conclusions about the potential of DPIEM for these applications, and some conclusions regarding the integrated operation of distributed enterprises/organizations with parallelism. The results of the statistical tests are shown in Tables 18 to 20.

The results from the statistical tests performed in the previous section can be interpreted as follows relative to conclusions about the advantages and disadvantages of the DPIEM application for distributed production organizations with parallelism.

(a) *Ability of DPIEM to find the minimum Φ .* DPIEM finds the minimum Φ when applied to the case studied. In the case, Φ for the base scenario is greater

than Φ for all scenarios with statistical significance as shown in Table 18. When compared to Φ^d of the base scenario, the solution proposed by DPIEM results in a maximum 82% lower Φ (Table 16). All scenarios recommended by the DPIEM model and tested in the TIE simulation environment resulted in cost savings relative to the base scenario.

(b) *Communication cost.* The DPIEM proposed solution resulting in the largest simulated variation relative to Φ^d , results also in low simulated Γ costs. This result is counter-intuitive, and apparently contradicts the expected trade-off behavior between Π and Γ costs. However, it must be noted that Π and Γ in Table 16 are accounted for on a server basis isolating the effect of a large number of servers in the system. If the Π and Γ costs are considered only in cost units (\$), the situation reverses. A higher communication cost is observed for lower operational costs as expected. The minimum cost solution proposed by DPIEM results in a 126% higher communication cost.

(c) *Reduced processing costs.* In all scenarios, Π decreases relative to the base scenario. The highest reduction in Π is reported for scenarios 1 and 3, both with a 98% reduction relative to the base scenario.

4. Conclusions

The development of the DPIEM was presented. The DPIEM model was conceived for the integration and optimization of distributed tasks for assisting e-commerce operations. DPIEM recognizes the importance of the integration objectives, however, this objective depends heavily on the situation at hand,

Table 16. Summary of TIE simulation cost results for all scenarios (negative variation indicates a higher cost)

Scenario	Based on DPIEM?	Average Φ (\$/Server)	Variation relative to Φ^d	Average Π (\$/Server)	Variation relative to Φ^d	Average Γ (\$/Server)	Variation relative to Φ^d
Base 1	No	4.250	—	3.237	—	1.013	—
1	Yes	0.745	82%	0.060	98%	0.685	32%
2	Yes	0.859	80%	0.109	97%	0.750	26%
3	Yes	1.023	76%	0.056	98%	0.967	4%
4	Yes	1.004	76%	0.110	97%	0.895	12%
5	No	1.701	60%	0.074	98%	1.656	— 64%
6	No	1.899	55%	1.277	61%	0.622	39%
7	No	2.117	50%	1.364	58%	0.753	26%
8	Yes	2.574	39%	1.691	48%	0.883	13%

Table 17. DPIEM design metrics

Scenario	Φ	Π	A	Ψ
1	20.248	8.148	20.248	24
2	20.258	8.196	20.258	24
3	24.758	10.405	24.758	24
4	24.767	10.453	24.767	24
7	21.612	9.162	21.612	6
8	161.176	75.380	161.176	4

and it may change during the integration process, shifting the attention to different variables and aspects of the distributed organizations. In consequence, the development of DPIEM considered this objective as external to the model. For the integration drivers, DPIEM uses the tasks being performed at the distributed organizations. The selection of tasks as the integration drivers, allow us to look for functional similarities across the distributed organizations for generating integration scenarios. An additional benefit to the utilization of tasks for integrating distributed organizations is the availability of graphical methods for representing the distributed tasks and their integration.

Based on previous research (Ceroni and Nof, 1997) server-parallelism exploitation was selected as the optimization variable to utilize in DPIEM. However,

Table 18. Statistical tests on average Φ from the simulation results (8 degrees of freedom, significance level = 5%, μ_i = average Φ for scenario i)

Statistical test on means ($t_{\text{critical}} = 2.306$)			
Test	$t_{\text{statistic}}$	$P(T \leq t)$	Result of test
$\mu_{bc1} = \mu_1$	7.6514	6.01×10^{-5}	$\mu_{bc1} > \mu_1$
$\mu_{bc1} = \mu_2$	7.1401	9.8×10^{-5}	$\mu_{bc1} > \mu_2$
$\mu_{bc1} = \mu_3$	7.1259	9.94×10^{-5}	$\mu_{bc1} > \mu_3$
$\mu_{bc1} = \mu_4$	7.1753	9.47×10^{-5}	$\mu_{bc1} > \mu_4$
$\mu_{bc1} = \mu_5$	-4.2724	0.0027	$\mu_{bc1} > \mu_5$
$\mu_{bc1} = \mu_6$	4.9734	0.001	$\mu_{bc1} > \mu_6$
$\mu_{bc1} = \mu_7$	4.4776	0.002	$\mu_{bc1} > \mu_7$
$\mu_{bc1} = \mu_8$	3.5108	0.007	$\mu_{bc1} > \mu_8$
$\mu_1 = \mu_2$	-0.6973	0.505	Non conclusive
$\mu_3 = \mu_4$	0.6717	0.520	Non conclusive
$\mu_1 = \mu_3$	-3.5840	0.007	$\mu_3 > \mu_1$
$\mu_1 = \mu_5$	2.4120	0.0423	$\mu_5 > \mu_1$
$\mu_2 = \mu_4$	-1.0002	0.346	Non conclusive
$\mu_7 = \mu_8$	-2.13068	0.065	Non conclusive
$\mu_6 = \mu_8$	-3.2741	0.011	$\mu_8 > \mu_6$

experiences in the literature use almost exclusively time-based performance measurements when increasing the number of parallel servers in the system. A major drawback of time-based performance measurements in engineering systems is that they do

Table 19. Analysis of variance for the serial content and number of sites (Degrees of freedom: Factors = 1, Interaction = 1, Within = 16, Significance level = 5%)

ANOVA two factors: Serial content of tasks and Number of sites						
Number of sites				$F_{\text{critical}} = 4.4939$		
		2	4	Factor	$F_{\text{statistic}}$	P-value
Serial	20%	Scenario 1	Scenario 3	Serial	0.3314	0.572
	60%	Scenario 2	Scenario 4	Sites	6.5836	0.020
				Interaction	0.6392	0.435

Result of test: Only the number of sites has statistically significant effect on the total cost.

Table 20. Analysis of variance for scenarios of experiment (Degrees of freedom: Between = 8, Within = 36, Significance level = 5%)

ANOVA one factor: scenarios base 1 to scenario 8 ($F_{\text{critical}} = 2.2085$):		
Factor	$F_{\text{statistic}}$	P-value
Between groups	24.9859	1.68×10^{-12}

Result of the test: There is statistically significant difference in observed Φ for the scenarios.

not consider the cost for the parallel servers being added to the system operation. This may result in unreal design recommendations and serious limitations for the method applicability. Therefore, to the server parallelism as optimization variable, the system operational cost (including server cost) was added as the performance measurement to optimize.

The optimization model devised operates over the integrated model of the distributed organizations. The integrated model of the distributed tasks is a logical model and it isolates implementation aspects of the tasks (location for example). This logical nature of the integrated model makes necessary to evaluate the cost the distributed assignment of the integrated tasks may generate. A special situation occurs when this cost results higher than the benefits from integrating and optimizing the distributed organizations. This situation forces the consideration and re-evaluation of the values for the cost parameters of the DPIEM model (execution and communication costs), and we need to repeat the application of DPIEM.

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References

- Armstrong, A. and Hagel III, J. (1996) The real value of on-line communities. *Harvard Business Review*, **74**(3), 134–141.
- Bernstein, A. J. (1966) Analysis of programs for parallel processing. *IEE Transactions on Electronic Computers*, **1**(EC-15), 757–763.
- Ceroni, J. (1999) *Models for Integration with Parallelism of Distributed Organizations*, Ph.D. Thesis, Purdue University.
- Ceroni, J., Nof, S. Y. and Matsui, M. (1999) Communication-based coordination modeling in distributed manufacturing systems. *International Journal of Production Economics*, **60–61**, 29–34.
- Ceroni, J. and Nof, S. Y. (1997) Planning Effective Parallelism in Production Operations, Research Memorandum No. 97-10, School of Industrial Engineering, Purdue University.
- Ceroni, J. (1996) *A Framework for Trade-Off Analysis of Parallelism*, MSIE Thesis, Purdue University.
- CIMOSA (1992), *Formal Reference Base*, version 2.0 ESPRIT Project Consortium AMICE.
- Dickman, S. (1995) Catching Customers on the Web, Inc. Technology **17**(4), 56–60.
- Hirsch, B., Kuhlmann, T., Marciniak, Z. K. and Maßow, C. (1995) Information system concept for the management of distributed production. *Computers in Industry*, **26**, 229–241.
- Khanna, N., Fortes, J. A. B. and Nof, S. Y. (1998) A formalism to structure and parallelize the integration of cooperative engineering design tasks. *IIE Transactions*, **30**, 1–15.
- Khanna, N. and Nof, S. Y. (1994) TIE: Teamwork Integration Evaluation Simulator: A Preliminary User Manual for TIE 1.1, Research Memorandum No. 94-21, School of Industrial Engineering, Purdue University.
- Kusiak, A. and Wang, J. (1995) Dependency analysis in constraint negotiation. *IEEE Transactions on Systems, Man, and Cybernetics*, **25**(9), 1301–1313.
- Lin, C. C. and Prasanna, V. K. (1995) Analysis of cost of performing communications using various communication mechanisms. *5th Symposium on the Frontiers of Massively Parallel Computation*, McLean, Va. pp. 290–297.
- McKichan, A. (1990) Retailing and computing—A happy union. *Business Quarterly*, **55**(1), 95–98.
- Mohd-Saman, M. Y. and Evans, D. J. (1995) Interprocedural analysis for parallel computing. *Parallel Computing*, **21**, 315–338.
- Nof, S. Y., Wilhelm, W. E. and Warnecke, H. J. (1997) *Industrial Assembly*, Chapman & Hall, London.
- Nof, S. Y. (ed.), (1994) *Information and Collaboration Models of Integration*, Kluwer Academic Publishers, Dordrecht.
- Papastavrou, J. D. and Nof, S. Y. (1992) Decision integration fundamentals in distributed manufacturing topologies. *IIE Transactions*, **24**(3), 27–42.
- Poirier, C. (1999) The convergence of business and technology. *Supply Chain Management Review*, **3**(3), 52–58.
- Qin, B., Sholl, H. A. and Ammar, R. A. (1991) Micro time cost analysis of parallel computations. *IEEE Transactions on Computers*, **40**, 613–628.
- Ramaswamy, S., Sapatnekar, S. and Banerjee, P. (1997) A

framework for exploiting task and data parallelism on distributed memory multicomputers. *IEEE Transactions on Parallel and Distributed Systems*, **8**(11), 1098–1116.

Xu, Z. and Hwang, K. (1996) Modeling communication overhead: MPI and MPL performance on the IBM SP2. *IEEE Parallel and Distributed Technology*, Spring, pp. 9–23.